The required test apparatus shall consist of either one two-celled chamber or two separate chambers in which the test conditions can be established and maintained. Chamber 1 shall be set to the operating low temperature and chamber 2 shall be set to the operating high temperature. The insertion of equipment in test chambers can modify its environmental condition. In this case, the temperature shall be stabilized within five minutes after transfer of the test item.

The following test procedure shall apply:

a. Place the equipment into chamber 1, and turn the equipment on.

b. Stabilize the equipment, if workable, at the operating low temperature.

c. Transfer the equipment from chamber 1 to chamber 2 as rapidly as possible but in no more than five minutes. In the case of when separate chambers are used, it is permissible to turn the equipment on immediately after the equipment transfer. If the transfer requires more than five minutes, it shall be indicated in the qualification form.

d. Stabilize the equipment at the operating high temperature, if workable. During this temperature stabilization, DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

e. Turn the equipment off, and keep it in a non operating state for a minimum of 2 minutes

f. Turn the equipment on, and transfer the equipment from chamber 2 to chamber 1 (from high to low temperature) as rapidly as possible but in no more than five minutes. In the case of when separate chambers are used, it is permissible to turn the equipment on immediately after the equipment transfer. If the transfer requires more than five minutes, it shall be indicated in the Qualification Form

g. Stabilize the equipment at the operating low temperature, if workable, and continue to operate the equipment for at least one hour. During the temperature stabilization, DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

h. Turn off the equipment for 30 minutes and restart the equipment while maintaining the chamber 1 at operating low temperature

i. Transfer the equipment at ambient temperature, and after equipment temperature stabilization, DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS in the second, or last, cycle as appropriate.

A minimum of two cycles (a. through i. above) shall be accomplished. If complete determination of compliance with applicable equipment performance standards can be accomplished during each temperature change period of a single cycle, then testing is required during the second cycle only. If the time during a temperature change period does not allow for complete determination of compliance with applicable equipment performance standards, a sufficient number of cycles shall be accomplished so that complete compliance can be determined. When temperature rise induces a potential risk of condensation on the equipment under test, the humidity level of the air in the chamber should be controlled to eliminate this condensation.

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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 6

Humidity

Important Notice

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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6.0 Humidity

6.1 Purpose of the Test

This test determines the ability of the equipment to withstand either natural or induced humid atmospheres. The main adverse effects to be anticipated are:

a. Corrosion.
b. Change of equipment characteristics resulting from the absorption of humidity. For example:

- Mechanical (metals).
- Electrical (conductors and insulators).
- Chemical (hygroscopic elements).
- Thermal (insulators).

6.2 Equipment Categories

Category A - Standard Humidity Environment

The standard humidity environment ordinarily provides an adequate test environment for equipment intended for installation in civil aircraft, non-civil transport aircraft and other classes, within environmentally controlled compartments of aircraft in which the severe humidity environment is not normally encountered.

Category B - Severe Humidity Environment

Equipment installed in zones not environmentally controlled may be required to be operated under conditions such that it is subjected to a more severe atmospheric humidity environment for periods of time in excess of that specified for the standard humidity environment.

Category C - External Humidity Environment

Equipment may be required to be operated under conditions such that it is subjected to direct contact with outside air for periods of time in excess of that specified for the standard humidity environment.

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6.3 Test Procedures

Subject the equipment to an atmosphere in which the relative humidity (RH) is 95±4 percent (%) RH, unless stated otherwise in the following steps. Moisture shall be provided by steam or by evaporation of water having a percent (pH) value between 6.5 and 7.5 or the water resistivity shall not be less than 250,000 ohm centimeters when measured at 25 degrees Celsius (°C). The velocity of air throughout the exposure area shall be between 0.5 and 1.7 meters per second. The test chamber shall be vented to the atmosphere to prevent buildup of pressure, and provisions shall be made to prevent water from dripping onto the equipment.

6.3.1 Category A—Standard Humidity Environment

The test profile is shown graphically in Figure 6-1. The procedure shall be in accordance with the following steps:

Step 1: Install the test item in the test chamber, and ensure its configuration is representative of that used in actual service.

Step 2: Stabilize the test item at 38±2 °C and 85±4 % RH.

Step 3: Over a two-hour period, ±10 minutes, raise the chamber temperature to 50±2 °C and humidity to 95±4 % RH.

Step 4: Maintain the chamber temperature at 50±2 °C and humidity at 95±4 % RH for six hours minimum.

Step 5: During the next 16-hour period, ±15 minutes, decrease the temperature gradually to 38±2 °C or lower. During this period, keep the humidity as high as possible and do not allow it to fall below 85 % RH.

Step 6: Steps 3, 4 and 5 constitute a cycle. Repeat these steps until a total of two cycles (48 hours of exposure) have been completed.

Step 7: At the end of the exposure period, remove the equipment from the test chamber and drain off (do not wipe) any condensed moisture. Within one hour after the two cycles are completed, apply normal supply power and turn on the equipment. Allow 15 minutes maximum following the application of primary power for the equipment to warm up. For equipment that does not require electrical power for operation, warm up the equipment for 15 minutes maximum by the application of heat not to exceed the short-time operating high temperature test as required by applicable equipment categories. Immediately following the warm-up period, make such tests and measurements as are necessary to DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

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6.3.2 Category B—Severe Humidity Environment

The test profile is shown graphically in Figure 6-2. The procedure shall be in accordance with the following steps:

Step 1: Install the test item in the test chamber, and ensure its configuration is representative of that used in actual service.

Step 2: Stabilize the test item at 38±2 °C and 85±4 % RH.

Step 3: Over a two-hour period, ± 10 minutes, raise the chamber temperature to 65±2 °C and humidity to 95±4 % RH.

Step 4: Maintain the chamber temperature at 65±2 °C and humidity at 95±4 % RH for six hours minimum.

Step 5: During the next 16-hour period, ±15 minutes, decrease the temperature gradually to 38±2 °C or lower. During this period, keep the humidity as high as possible and do not allow it to fall below 85 % RH.

Step 6: Steps 3, 4 and 5 constitute a cycle. Repeat these steps until a total of 10 cycles (240 hours of exposure) have been completed.

Step 7: At the end of the exposure period, remove the equipment from the test chamber and drain off (do not wipe) any condensed moisture. Within one hour after the 10 cycles are completed, apply normal supply power and turn on the equipment. Allow 15 minutes maximum following the application of primary power for the equipment to warm up. For equipment that does not require electrical power for operation, warm up the equipment for 15 minutes maximum by the application of heat not to exceed the short-time operating high temperature test as required by applicable equipment categories. Immediately following the warm-up period, make such tests and measurements as are necessary to DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

6.3.3 Category C—External Humidity Environment

The test profile is shown graphically in Figure 6-3. The procedure shall be in accordance with the following steps:

Step 1: Install the test item in the test chamber, and ensure its configuration is representative of that used in actual service.

Step 2: Stabilize the test item at 38±2 °C and 85±4 % RH.

Step 3: Over a two-hour period, ±10 minutes, raise the chamber temperature to 55±2 °C and humidity to 95±4 % RH.

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Step 4: Maintain the chamber temperature at 55±2 °C and humidity at 95±4 % RH for six hours minimum.

Step 5: During the next 16-hour period, ±15 minutes, decrease the temperature gradually to 38±2 °C or lower. During this period, keep the humidity as high as possible and do not allow it to fall below 85 %.

Step 6: Steps 3, 4 and 5 constitute a cycle. Repeat these steps until a total of six cycles (144 hours of exposure) have been completed.

Step 7: At the end of the exposure period, remove the equipment from the test chamber and drain off (do not wipe) any condensed moisture. Within one hour after the six cycles are completed, apply normal supply power and turn on the equipment. Allow 15 minutes maximum following the application of primary power for the equipment to warm up. For equipment that does not require electrical power for operation, warm up the equipment for 15 minutes maximum by the application of heat not to exceed the short-time operating high temperature test as required by applicable equipment categories. Immediately following the warm-up period, make such tests and measurements as are necessary to DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

6.3.4 Conducting Spot Checks

For conducting spot checks on the performance of the equipment under test, the equipment may be operated at the end of each of the 6 or 10 cycles as appropriate for a period not to exceed 15 minutes. If the equipment is removed from the test chamber to conduct a spot check, the period of removal shall not exceed 20 minutes, and the equipment shall not be operated for more than 15 minutes of this 20-minute period.

6.3.5 Other Specified Checks

If the applicable performance standard requires that other checks be made to determine compliance, these shall also be performed during this test.

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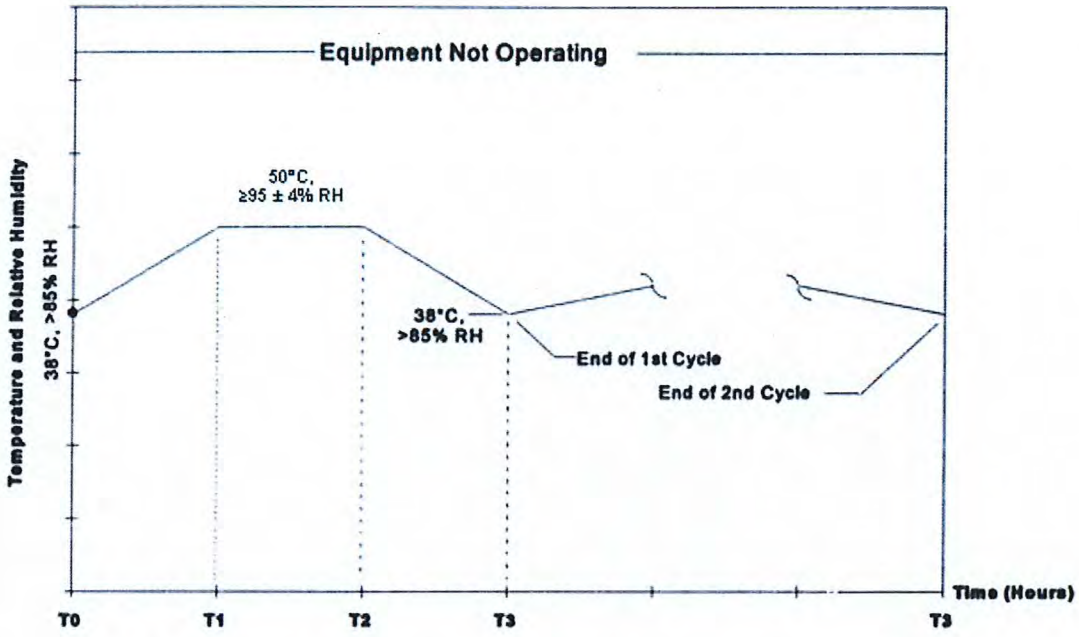

Figure 6-1 Category A - Standard Humidity Environment Test

NOTES:

1) T0 to T1 is 2 hours ±10 minutes.
2) T1 to T2 is 6 hours, minimum.
3) T2 to T3 is 16 hours ±15 minutes. During this period, relative humidity should not fall below 85%.
4) See paragraph 6.3.1, Step 7, for continuation of test after the end of the 2nd cycle.
5) T0 represents the beginning of the subsequent cycle(s), not the initial point of the test.

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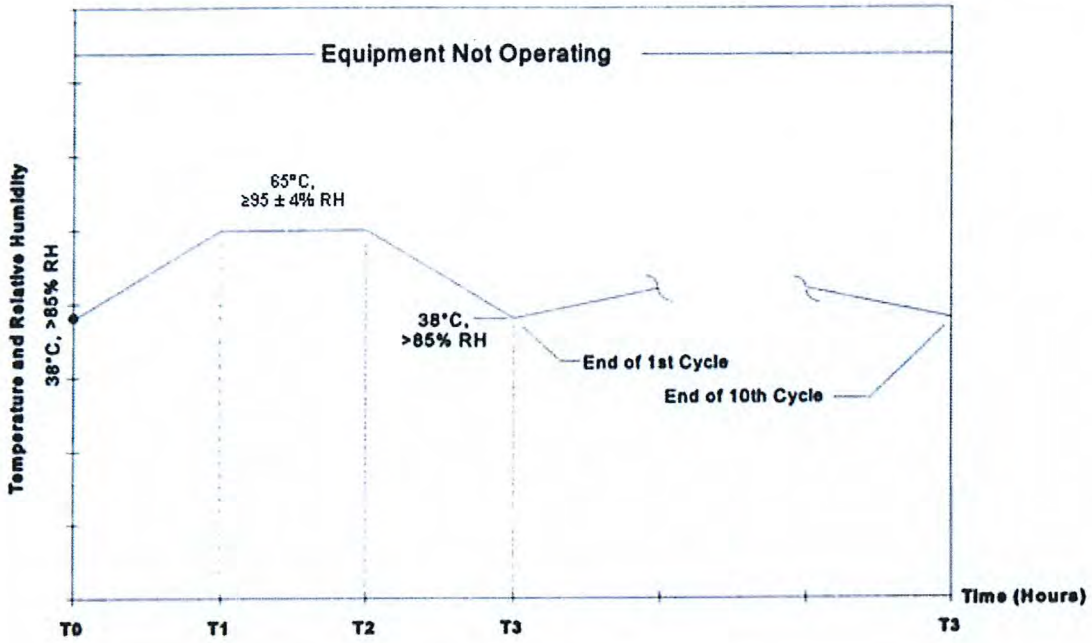

Figure 6-2 Category B - Severe Humidity Environment Test

NOTES:

1) T0 to T1 is 2 hours ±10 minutes.
2) T1 to T2 is 6 hours, minimum.
3) T2 to T3 is 16 hours ±15 minutes. During this period, relative humidity should not fall below 85%.
4) See paragraph 6.3.2, Step 7, for continuation of test after the end of the 10th cycle.
5) T0 represents the beginning of the subsequent cycle(s), not the initial point of the test.

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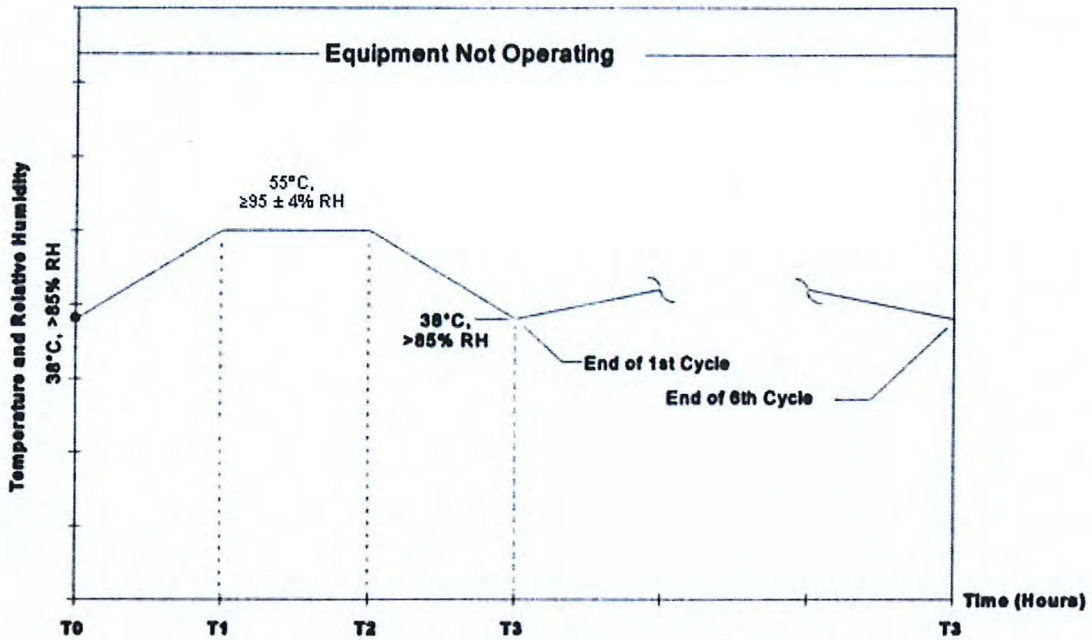

Figure 6-3 Category C - External Humidity Environment Test

NOTES:

1) T0 to T1 is 2 hours ±10 minutes.
2) T1 to T2 is 6 hours, minimum.
3) T2 to T3 is 16 hours ±15 minutes. During this period, relative humidity should not fall below 85%.
4) See paragraph 6.3.3, Step 7, for continuation of test after the end of the 6th cycle.
5) T0 represents the beginning of the subsequent cycle(s), not the initial point of the test.

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